

Ex No: 1b

IMPLEMENTATION OF DEPTH FIRST SEARCH

Aim:

To implement depth first search

Procedure:

Define the Graph Class: Create a Graph class initialized with a specified number of vertices (self.vertices) and an adjacency list representation (self.graph) using a dictionary.

Add Edges: Implement the add_edge method to establish undirected edges between vertices u and v by appending each to the other's adjacency list

Initialize Visited Array: In the main function, initialize a boolean list (visited) with False values corresponding to the number of vertices to keep track of visited nodes.

Define Recursive DFS: Implement the dfs(v, visited) recursive function:

Print the current vertex v.

- Mark the current vertex \$v\$ as visited (visited[v] = True).
- Iterate through all adjacent neighbors of vertex \$v\$.
- For each unvisited neighbor (if not visited[neighbor]), recursively invoke dfs(neighbor, visited)

Execute Search: Instantiate the graph with 6 vertices, add the defined edges, and start the traversal from the source vertex 0.

Program:

```
# DFS Algorithm Implementation
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```
class Graph:
```

```
    def __init__(self, vertices):
```

```
        self.vertices = vertices
```

```
        self.graph = {i: [] for i in range(vertices)}
```

```
# Function to add an edge to the graph
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```
def add_edge(self, u, v):
```

```
self.graph[u].append(v)
self.graph[v].append(u)

# Recursive function for DFS traversal
def dfs(self, v, visited):

    print(v, end=" ") # Print the current node
    visited[v] = True # Mark the node as visited

    # Recur for all the adjacent vertices
    for neighbor in self.graph[v]:

        if not visited[neighbor]:

            self.dfs(neighbor, visited)

# Main function to demonstrate DFS
def depth_first_search():

    # Creating a graph with 6 vertices
    g = Graph(6)

    # Adding edges
    g.add_edge(0, 1)
    g.add_edge(0, 2)
    g.add_edge(1, 3)
    g.add_edge(1, 4)
    g.add_edge(2, 5)

    # To track visited nodes
    visited = [False] * g.vertices

    # Start DFS traversal from vertex 0
    print("Depth First Search starting from vertex 0:")
    g.dfs(0, visited)
```

```
# Calling the main function  
depth_first_search()
```

Output:

```
Depth First Search starting from vertex 0:  
0 1 3 4 2 5
```